held disproportionately by a few investors. Investors have often resorted to *factor replication* to track fixed income indexes.[54] They track fixed income indexes well despite owning only a fraction of the bonds because factors dominate, especially the level factor, in fixed income portfolios.

6.5. SUMMARY

Rarely should an investor base his holdings of risk-free, government bonds on market capitalization weights or outstanding issue sizes. Safe assets collect many different sources of factor risk, many at odds with one another. The factor investing approach starts with listing the factors that are most important for an asset owner and reflecting these factors in the optimal country weights.

7. Passive-Aggressive Norway Redux

As part of the active management review in 2009, NBIM wrote a letter to the Ministry of Finance stating:[55] "Active management will expose the Fund to systematic risk factors to a greater or lesser extent. The management and control of systematic risk must therefore be part of our management task." NBIM acknowledged the importance of factors but wanted them to be under its control. This is an important debate about where the factor decision should be made. In another letter dated February 2, 2012, to the Ministry of Finance, NBIM wrote:

> The investment strategy should be designed in such a way that the Fund can harvest risk premia dynamically. . . . Norges Bank believes that the strategic benchmark index should not be adjusted to take account of systematic risk premia for equity investments.

As I write this, the Ministry of Finance has not made a final recommendation on whether the dynamic factor decision should be at the asset owner level (Parliament) or at the fund manager level (NBIM). (The issue was discussed in the Ministry's 2013 report to Parliament.) Nevertheless, NBIM has gone ahead and introduced factors into its internal benchmarks. In 2012, NBIM introduced an *operational benchmark* for equities that explicitly takes into account systematic risk factors. There are just two factors so far: value-growth and size. (Note that I believe size by itself does not yield a risk premium in excess of market exposure.) I anticipate there will be reporting of exposures to the systematic risk factors so that the ultimate owners of the fund, the Norwegian people, are informed of the

[54] The quality of many fixed income indexes is far below equity indexes. See Goltz and Campani (2011).

[55] Letter to the Ministry of Finance dated December 23, 2009.